Assessment of Clinical Competencies in Medical Education in the Era of Artificial Intelligence: An Integrative Literature Review

Abstract

Artificial intelligence (AI) has rapidly expanded within medical education, challenging traditional approaches to assessing clinical competence. Competency-based curricula require assessment strategies capable of inferring professional performance, clinical reasoning, communication, and decision-making in authentic clinical contexts. However, written examinations and case-based assignments, historically used as proxies for competence, are increasingly affected by AI systems capable of generating clinically structured responses. This transformation raises concerns regarding the validity of assessment evidence and the interpretation of learner performance. This integrative literature review aimed to analyze how clinical competence assessment in medical education has been discussed in the context of AI. Searches were conducted in PubMed, Scopus, Web of Science, and ERIC, including studies published between 2019 and 2026. Twenty-four studies met the inclusion criteria. Data were synthesized through interpretive thematic analysis and categorized according to levels of evidence. The findings indicate that AI does not replace the human assessor but contributes to a hybrid assessment ecosystem. Performance-based methods, such as OSCEs, workplace-based assessments, and portfolios, gain relevance, while assessment shifts

from verifying correct answers to evaluating reasoning processes and justification. AI-mediated feedback enhances individualized learning and longitudinal monitoring; however, risks include superficial performance, compromised authenticity, and challenges to professional autonomy. Ethical concerns involve authorship transparency, reliability, and patient safety. Overall, clinical competence assessment must evolve from knowledge recall toward evaluation of judgment in human-AI interaction. Future assessment models should explicitly incorporate critical use of AI as part of contemporary medical competence while preserving the centrality of clinical reasoning and structured faculty mediation.